

Adeno-associated virus 2 (AAV2) is a nonpathogenic, nonenveloped DNA virus that requires helper viruses such as adenovirus (AV) or herpes simplex virus (HSV) to replicate efficiently. Because it can transduce human cells, including quiescent (nondividing) cells, without causing disease, AAV2 is attractive as a gene therapy vector. In clinical trials, a modified AAV2 virus has successfully treated the retinal disease Leber's congenital amaurosis as well as the blood clotting disorder hemophilia B. AAV2 transduces cells via a multistep process, shown in Figure 1.

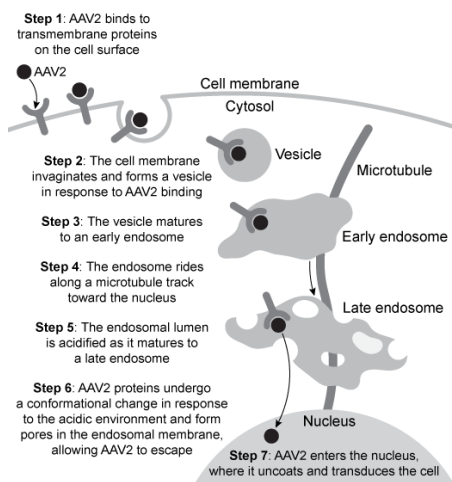


Figure 1 Schematic showing the mechanism of transduction by AAV2

Studies indicate that unlike AV and HSV, AAV2 inhibits pathways other than the endosomal pathway, such as trafficking to the Golgi, increasing the percentage of invading virus particles that arrive at the nucleus.

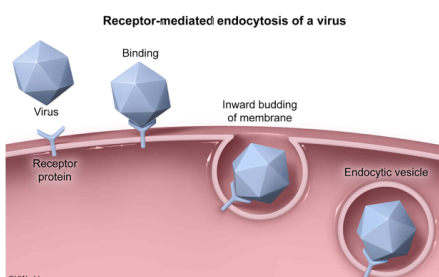
AAV2 entry into cells could best be blocked by a drug that inhibits which of the following mechanisms?

- ☐ A. Fusion of viral and cell membranes upon receptor binding [21%]
- ☐ B. Formation of membrane extensions that engulf external particles [10%]
- ☒ C. Inward budding of the cell in response to receptor binding [61%]
- ☐ D. Constitutive invagination of the cell membrane to take up extracellular liquid [5%]

Correct

60% answered correctly

Explanation:



Cells constantly take up matter from their surrounding environment by **endocytosis**, in which the surroundings are enveloped by a portion of the cell membrane. That portion subsequently pinches off from the rest of the membrane and becomes a vesicle inside the cell. This vesicle then matures and merges with other membrane-bound compartments to become an **endosome**. Cells can conduct endocytosis through **three major mechanisms**:

phagocytosis, pinocytosis, and receptor-mediated endocytosis.

Figure 1 shows that AAV2 enters cells by **receptor-mediated endocytosis**, in which an external ligand (ie, AAV2) binds specific receptor proteins on the cell surface. This binding induces the plasma membrane to bud inward toward the cytosol before pinching off as a vesicle that contains both the ligand and its receptor. Therefore, a drug that inhibits this inward budding would likely prevent AAV2 from entering cells.

(Choice A) Viruses with a capsid encased in a phospholipid membrane, known as enveloped viruses, can **fuse their membranes** with the cell membrane. When this occurs, no vesicle is formed, and the capsid is released directly into the cytosol. As a nonenveloped virus, AAV2 cannot enter cells by this method; therefore, membrane fusion is not the method of entry shown in Figure 1.

(Choice B) **Phagocytosis**, also called "cell eating," occurs when cells engulf external particles by extending portions of their membrane to reach around the particle rather than by the inward budding of the membrane, shown in Figure 1.

(Choice D) **Pinocytosis** is a nonspecific process (ie, does not require receptor binding) that involves the continuous invagination of the cell membrane to take up extracellular fluid. Because it is a distinct process, inhibiting pinocytosis does not inhibit receptor-mediated endocytosis.

Educational objective:

Cells take up their surrounding environment via endocytosis, which includes mechanisms of phagocytosis, pinocytosis, and receptor-mediated endocytosis. Many viruses use receptor-mediated endocytosis to enter cells. Enveloped viruses can also enter cells by fusing their membrane with the membrane of the cell.

Time spent:0 sec

Subject:
Biology

QID:401598

System:
Cellular Biology

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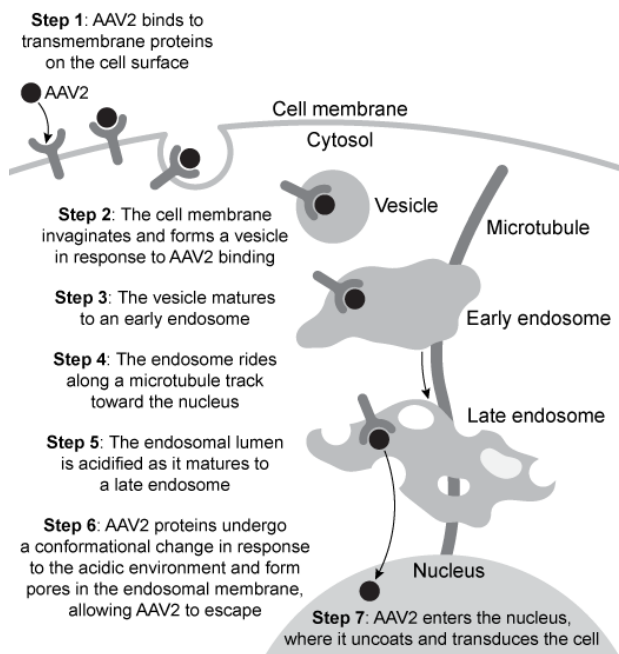


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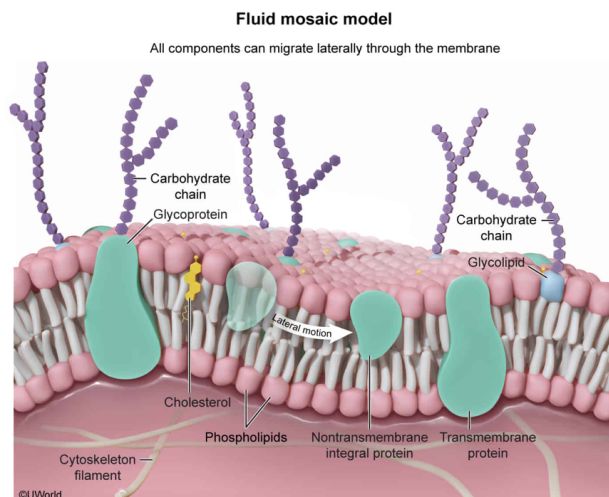
Which of the following is true concerning the proteins to which AAV2 binds in Figure 1, Step 1?

- ☐ A. They localize primarily in low-cholesterol environments known as lipid rafts. [9%]
- ☐ B. They are held statically in place by cytoskeletal components. [13%]
- ☐ C. They are able to interact only with the phospholipids on the outer leaflet of the cell membrane. [14%]
- ☒ D. They can migrate laterally through the phospholipid-rich environment of the cell membrane. [62%]

Correct

62% answered correctly

Explanation:



The membrane of an animal cell is described by the **fluid mosaic model**, in which transmembrane proteins and other components reside in a **fluid bilayer** composed largely of **phospholipids**. The cell membrane is said to be **fluid** as its various nonphospholipid components **can migrate laterally** (ie, from side to side) through the entire phospholipid-rich surface of the cell in any direction. These other important components include **cholesterol**, **glycoproteins**, and **glycolipids** (proteins and lipids that have been modified with carbohydrates).

Many membrane proteins, called **transmembrane proteins**, span the width of the membrane. Figure 1 shows that AAV2 binds to transmembrane proteins, which must be able to migrate laterally through the phospholipid environment of the cell membrane.

(Choice A) Although it is believed that transmembrane proteins often localize to relatively ordered portions of the membrane called **lipid rafts**, these rafts are *high* in cholesterol content. In addition, the passage gives no information about the structure of the AAV2 transmembrane receptor, so specifics of its location within the cell membrane cannot be determined.

(Choice B) Cytoskeletal components such as actin filaments interact with some transmembrane proteins, but they do not hold these proteins statically in place. Therefore, these transmembrane proteins can move dynamically within the cell membrane.

(Choice C) The phospholipid bilayer of the cell membrane has an outer leaflet that interacts with the extracellular environment and an inner leaflet that interacts with the cytoplasm of the cell. *Transmembrane* proteins cross the entire phospholipid bilayer and therefore interact with both the outer and inner leaflets. Proteins that interact with only one leaflet are not transmembrane proteins.

Educational objective:

Cell membranes are composed largely of phospholipids, which act as a fluid that allows other membrane components such as transmembrane proteins, glycoproteins, cholesterol, and glycolipids to migrate through this environment laterally. Accordingly, the structure of the animal cell membrane is known as the fluid mosaic model.

Time spent:0 sec**Subject:**
Biology**QID:**401599**System:**
Cellular Biology

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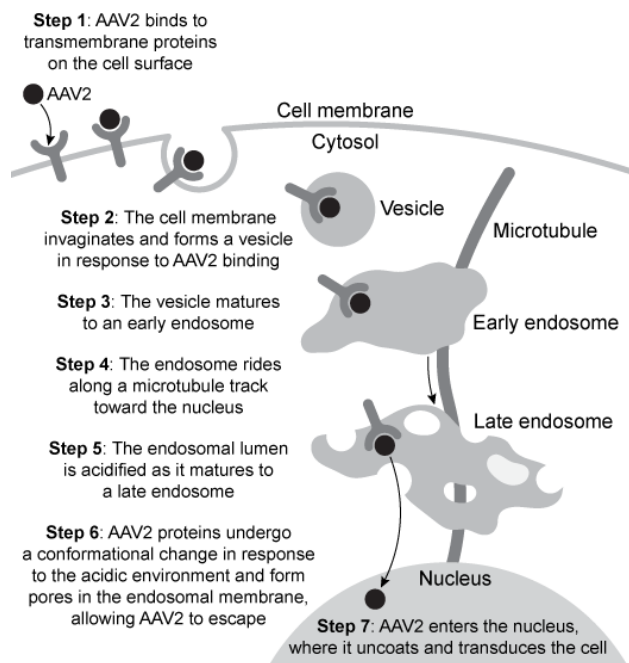


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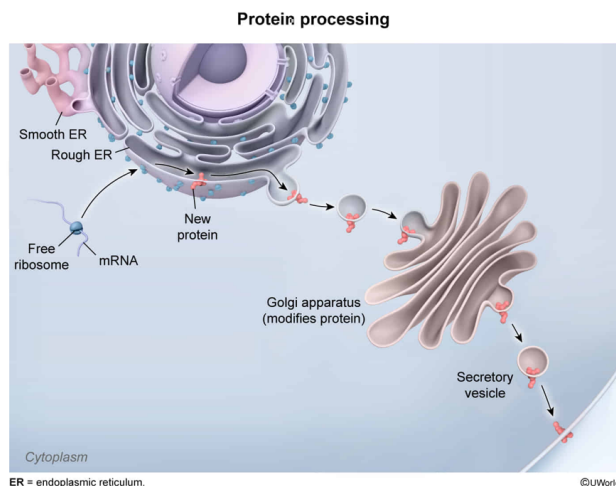
Prior to infection by AAV2, the receptors that bind to AAV2 must reach the cell surface. Along the path to the cell surface, these receptors must pass through all of the following cellular compartments EXCEPT:

- ☐ A. the endoplasmic reticulum. [2%]
- ☐ B. the Golgi apparatus. [3%]
- ✓ ☒ C. a peroxisome. [88%]
- ☐ D. a secretory vesicle. [5%]

Correct

87% answered correctly

Explanation:



Proteins that are **embedded in the plasma membrane** traveled through the **secretory pathway** to arrive there. Translation of mRNA to proteins always begins in the cytosol, but proteins intended for the secretory pathway (eg, the receptors to which viruses such as AAV2 bind) have an N-terminal sequence called a **signal sequence**. Once the signal sequence is recognized, the ribosome is transported to the **rough endoplasmic reticulum (RER) (Choice A)**, where the protein is either translocated across the RER membrane (secreted proteins) or embedded into the membrane (integral membrane proteins).

In the RER, some **post-translational modifications** may occur, including glycosylation, disulfide bond formation, phosphorylation, and protein cleavage. The RER then packages proteins into vesicles and sends them to the **Golgi apparatus (Choice B)**, where they are further processed. Finally, the Golgi packages proteins destined for the plasma membrane into **secretory vesicles (Choice D)**, which fuse with the plasma membrane. Proteins that were initially embedded in the RER membrane frequently end up embedded in the plasma membrane as receptor proteins.

The **peroxisome** is not part of the secretory pathway and is **not involved in protein transport** to the cell surface. Instead, peroxisomes contain various **oxidoreductase enzymes** and function to help maintain the proper oxidation state within cells. Proteins in the peroxisomal lumen were translated entirely in the cytosol and imported. Therefore, the proteins that bind AAV2 most likely do not pass through the peroxisome on their way to the cell surface.

Educational objective:

The secretory pathway involves the processing of proteins as they go through the endoplasmic reticulum and the Golgi apparatus. After processing in the Golgi, proteins destined for the cell membrane are packed into secretory vesicles, which

fuse with the plasma membrane. Other organelles, such as the nucleus, mitochondria, and peroxisomes, are not involved in the secretory pathway, and most of their proteins are translated in the cytosol.

Time spent:0 sec

Subject:

Biology

QID:401600

System:

Cellular Biology

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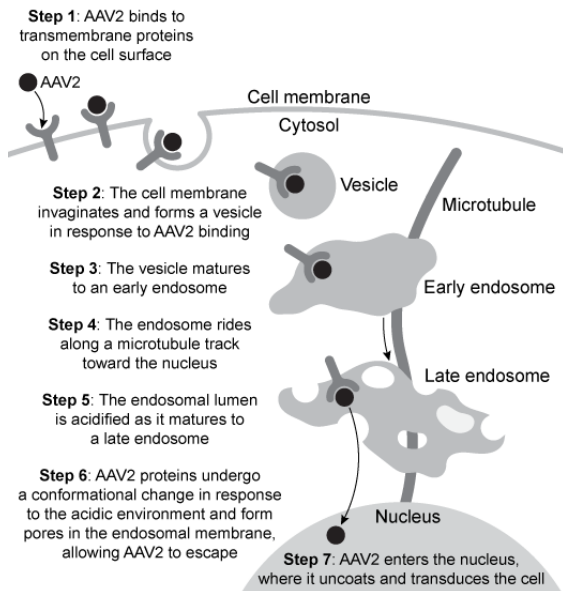


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As it matures, the endosomal lumen becomes acidified relative to the cytosol. Protons from the cytosol most likely enter the endosomal lumen by which of the following processes?

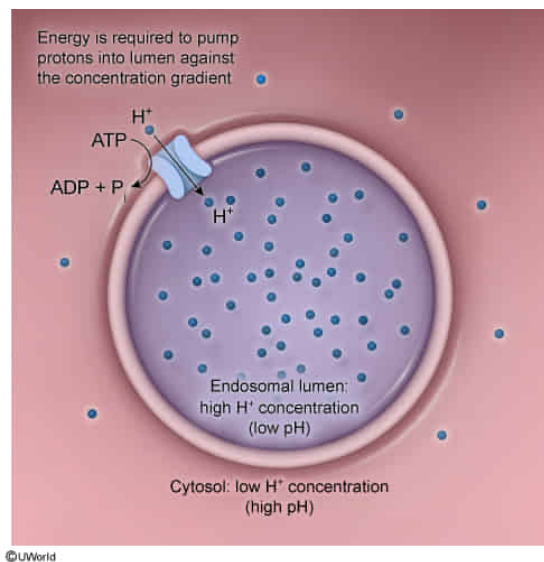
- ☐ A. Simple diffusion [14%]
- ☐ B. Facilitated diffusion [21%]
- ☐ C. Passive transport [8%]
- ✓ ☒ D. Active transport [54%]

Correct

54% answered correctly

Explanation:

Endosomal acidification requires active transport



In the absence of external energy input, molecules in solution diffuse from areas of **high concentration to low concentration**. Such flow down a concentration gradient is **thermodynamically favorable**. Molecules that are separated from an area of low concentration by a membrane will still flow across the membrane down their concentration gradient as long as the membrane is permeable to those molecules.

When molecules undergo net movement from a region of one concentration into a region of **greater or equal concentration**, they are moving **against their concentration gradient**. For this to occur, **energy** must be put **into the system** because the process is thermodynamically unfavorable. Transport systems that require energy input and move substances against their concentration gradient are **active transport systems**.

Endocytic vesicles obtain their fluid from the extracellular space, which is at physiological pH (7.4). This is the same as the pH of the cytosol. Consequently, acidification of the endosomal lumen involves the transport of protons against their concentration gradient from an area of relatively low concentration (cytosol; pH $\approx$ 7.4) to one of equal or higher concentration (endosomal lumen; pH $\approx$ 7.4-5.5). Therefore, protons from the cytosol can only **enter the endosomal lumen through active transport**.

(Choices A, B, and C) Simple diffusion and facilitated diffusion are both forms of **passive transport**, in which no energy input is required and molecules *diffuse down their concentration gradient*. In simple diffusion, molecules that are small or lipid-soluble travel across the cell membrane directly, without help from protein channels. In facilitated diffusion, molecules that cannot normally cross the membrane on their own due to charge or size are allowed to pass through protein channels or gates that do not require energy input to operate.

Educational objective:

Molecules can cross biological membranes by active or passive transport. Active transport involves a molecule moving against its concentration gradient and requires energy input (eg, ATP). In contrast, passive transport involves molecules diffusing down their concentration gradient and requires no external energy. Passive transport can be divided into simple diffusion, in which molecules cross membranes directly, and facilitated diffusion, in which molecules must go through protein channels.

Time spent:0 sec

Subject:

Biology

QID:401601

System:

Cellular Biology

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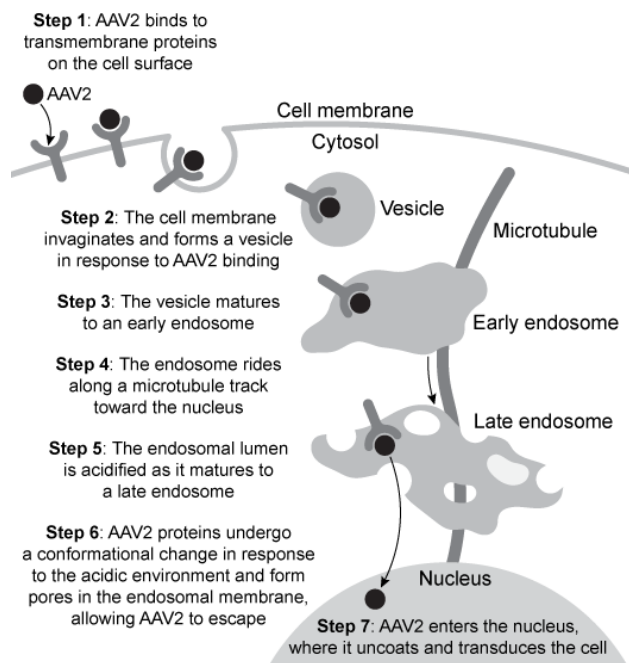


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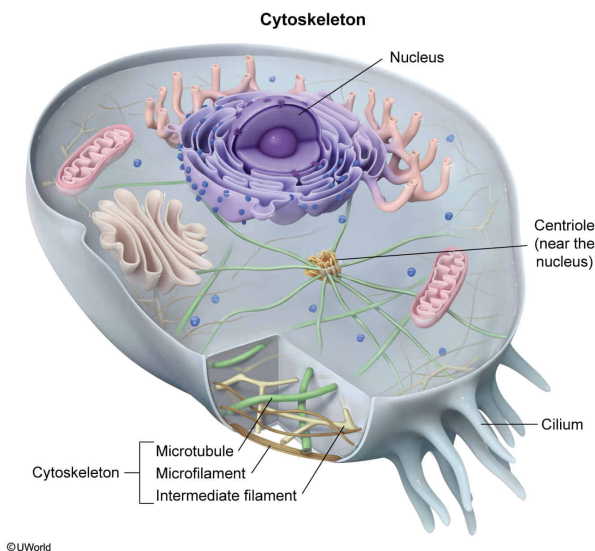
In quiescent cells, early endosomes containing AAV2 would most likely move toward:

- ☐ A. mitochondria. [37%]
- ✓ ☒ B. centrioles. [36%]
- ☐ C. cilia. [8%]
- ☐ D. kinetochores. [17%]

Correct

36% answered correctly

Explanation:



The **cytoskeleton** is an intracellular scaffolding network of fibers that are interspersed throughout the cytoplasm. It functions to give the cell its shape, support cellular motility, and help organize intracellular components. The three **major cytoskeletal components** are actin filaments (ie, microfilaments), intermediate filaments, and microtubules.

Microtubules are composed of alternating α - and β -tubulin subunits that assemble into hollow tubes. They are involved in various forms of movement within the cell; microtubules form the mitotic spindle during **cell division** and are essential for **cell motility** as they form the core of **cilia and flagella**. Microtubules also **facilitate transport of vesicles** such as endosomes and other organelles from one location in the cell to another. The microtubules involved in intracellular transport originate **near the nucleus** on small organelles called centrosomes. The centrosome consists of two smaller entities called **centrioles**, and microtubules radiate out from this organelle toward the plasma membrane.

The passage indicates that endosomes containing AAV2 move along microtubule tracks toward the nucleus. Because AAV2 enters the cell at the plasma membrane and moves along microtubules toward the nucleus, the endosomes that carry it must be moving toward the centrioles, which are located near the nucleus.

(Choice A) Mitochondria are organelles that carry out several metabolic processes. They are not generally connected to microtubules, so the endosomes do not travel toward mitochondria.

(Choice C) Microtubules are involved in cilia formation. However, cilia are found at the edge of the cell, and the endosomes are moving away from the edge and toward the nucleus in the central region of the cell.

(Choice D) A **kinetochore** is a protein-DNA complex to which spindle fibers, made of microtubules, attach during metaphase of mitosis. However, quiescent (nondividing) cells are in the G_0 phase of the **cell cycle** and therefore do not have organized kinetochores because the chromosomes are uncoiled and relatively unstructured in this phase.

Educational objective:

The cytoskeleton consists of actin filaments (microfilaments), intermediate filaments, and microtubules. Microtubules are involved in intracellular transport. The microtubules involved in this process originate near the nucleus on organelles called centrioles and radiate out toward the plasma membrane.

Time spent:0 sec

Subject:
Biology

QID:401602

System:
Cellular Biology

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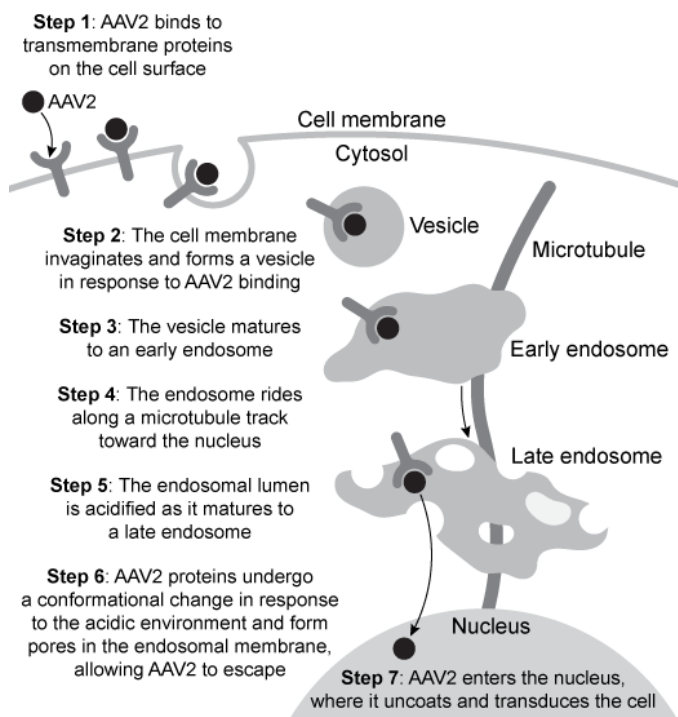


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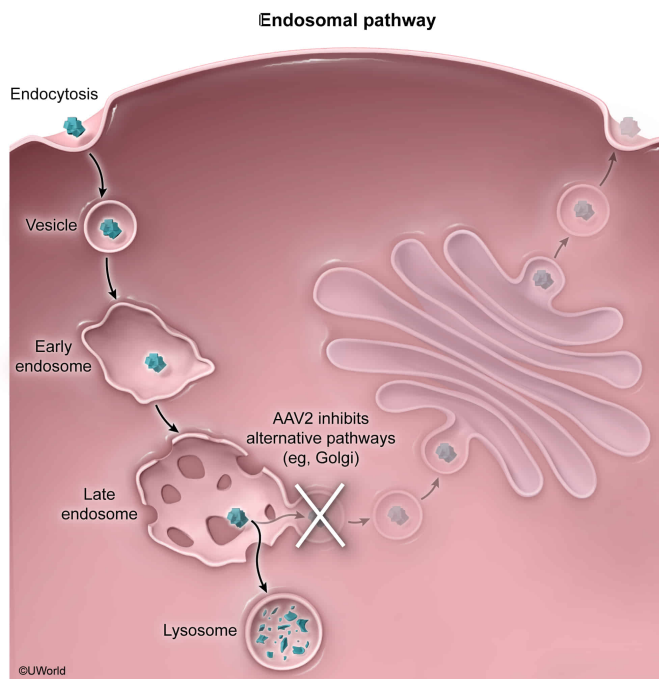
What is the fate of the virus particles that fail to escape the acidified endosome in Figure 1, Step 6?

- ✔ ☒ A. Degradation in the lysosome [76%]
- ☐ B. Repackaging in the Golgi followed by exocytosis [6%]
- ☐ C. Reprocessing in the endoplasmic reticulum [2%]
- ☐ D. Destruction by a proteasome [15%]

Correct

76% answered correctly

Explanation:



The **endosomal pathway** begins at endocytosis where, following internalization of extracellular materials into vesicles, the vesicles containing these materials mature to **early endosomes**, **late endosomes**, and finally, **lysosomes**. The lysosome is a membrane-enclosed organelle that serves as the "digestive system" of the cell; it contains an acidic environment ($\text{pH} \approx 4.5$) and various hydrolytic enzymes that facilitate the **degradation of various biomolecules** (eg, proteins, carbohydrates, nucleic acids, lipids).

Some endosomes participate in alternative pathways, such as trafficking, through the Golgi apparatus. This pathway allows the cargo within the vesicles to enter the secretory pathway and be expelled from the cell by exocytosis (ie, bypassing degradation by the lysosome).

The passage states that *AAV2 inhibits* pathways other than the endosomal pathway. Therefore, AAV2 particles that fail to escape acidified endosomes in Step 6 cannot enter the Golgi or other secretory pathway organelles to undergo exocytosis (**Choice B**). The only other available fate is the lysosome (the last step of the endosomal pathway), where AAV2 particles will be degraded.

(Choice C) Endosomes generally do not transfer their cargo to the endoplasmic reticulum (ER). However, if they did, this would still not be part of the endosomal pathway, as the ER is part of the **secretory pathway**. Per the passage, AAV2 inhibits such alternative pathways.

(Choice D) The **proteasome** is a protein complex found in the cytosol that typically degrades ubiquitinated proteins. For viral particles to follow this pathway, they would first have to escape the endosome and enter the cytosol, and particles that fail to escape will never interact with the proteasome.

Educational objective:

After internalization of extracellular materials into vesicles, the vesicles containing these materials become endosomes, which then mature from an early stage to a late stage, ultimately becoming lysosomes. Endocytic cargo that does not enter alternative pathways (eg, entry into the secretory pathway) and fails to escape from endosomes will end up in the lysosome to be degraded.

Time spent:0 sec

Subject:
Biology

QID:401603

System:
Cellular Biology